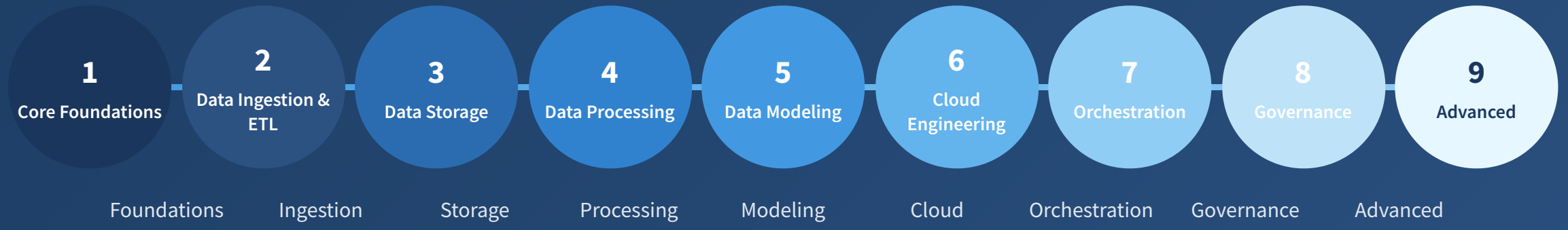
Data Engineering Roadmap

A Step-by-Step Guide to Mastering Data Engineering

Data Engineering Roadmap Overview

A visual journey through the 9 stages of mastering data engineering



1 Core Foundations

Building blocks for any data engineer

Skills & Concepts

- **ETL/ELT** Data Pipelines
- **Batch vs Streaming** Processing
- **Data Lakes vs** Data Warehouses
- **Relational:** MySQL, PostgreSQL
- **NoSQL:** MongoDB, Cassandra
- **Python** (primary)
- **SQL** (essential)

Tools

- **SQL clients:** DBeaver, pgAdmin
- **Python libraries:** pandas, numpy
- **IDEs:** VS Code, PyCharm
- **Version Control:** Git, GitHub

Resources

- **Book:** Fundamentals of Data Engineering
- **Practice:** SQLZoo, LeetCode SQL
- **Projects:** Build simple ETL pipelines

2 Data Ingestion and ETL

Moving and transforming data efficiently

Skills & Concepts

- **ETL vs ELT** approaches
- **Batch processing** with scheduled jobs
- **Streaming** data ingestion
- **Data quality** and validation

Tools

- **Apache Airflow** (workflow)
- **Apache NiFi** (data flow)
- **AWS Glue** (cloud ETL)
- **Kafka** (stream ingestion)
- **Spark** (processing)
- **DBT** (transformation)

Hands-on Projects

- **Build ETL pipeline** with Python + Airflow
- **Stream data** from Kafka to database
- **Transform & load** with DBT

3 Data Storage & Management

Storing and organizing data efficiently

Skills & Concepts

- **Data Warehouses** vs Data Lakes
- **Schema design** and evolution
- **Normalization** vs Denormalization
- **Partitioning** strategies
- **Indexing** and clustering
- **Metadata** and cataloging

Warehouses

- **Snowflake** - Cloud-native
- **BigQuery** - Google Cloud
- **Redshift** - AWS
- **Azure Synapse** - Microsoft

Lakes & Tools

- **AWS S3** - Object storage
- **GCS** - Google Cloud Storage
- **Hive / Presto / Trino** - Query engines
- **Delta Lake / Iceberg** - ACID on lakes

4 Data Processing & Transformation

Processing large datasets effectively

Skills & Concepts

- **Distributed computing** concepts
- **Batch processing** workflows
- **Stream processing** techniques
- **Data cleaning** methods
- **Data validation** strategies
- **Data enrichment** techniques

Core Tools

- **Apache Spark** (PySpark)
- **Apache Flink** (streaming)
- **Pandas** (small datasets)
- **Dask** (medium datasets)

Real-time Processing

- **Kafka Streams** (real-time)
- **Spark Streaming** (micro-batch)
- **Windowing** operations
- **State management** in streams

5 Data Modeling & Analytics

Making data usable for BI and analytics

Skills & Concepts

- **Dimensional modeling** (star & snowflake)
- **OLAP vs OLTP** systems
- **Data aggregation** techniques
- **Metrics calculation** methods
- **Slowly Changing Dimensions** (SCDs)

Transformation Tools

- **DBT** (data transformation)
- **SQL & advanced querying**
- **Window functions** & CTEs
- **Data modeling frameworks**

Visualization Tools

- **Looker** (Google)
- **Tableau** (interactive dashboards)
- **Power BI** (Microsoft)
- **Metric layer tools** (Superset, Atlan)

6 Cloud Data Engineering

Building modern pipelines in the cloud

Skills & Concepts

- **Cloud storage** fundamentals
- **Computing basics** & scaling
- **Security** & governance
- **Cost optimization** strategies
- **Managed services** integration

AWS

- **S3** - Object storage
- **Redshift** - Data warehouse
- **Glue** - ETL service
- **EMR** - Big data processing
- **Kinesis** - Streaming

GCP & Azure

- **BigQuery** - Data warehouse
- **Dataflow** - Stream/batch
- **Pub/Sub** - Messaging
- **Data Lake** - Azure storage
- **Synapse** - Analytics
- **Data Factory** - ETL

7 Orchestration, Scheduling & Monitoring

Keeping pipelines running smoothly

Skills & Concepts

- **Job scheduling** strategies
- **Workflow orchestration** patterns
- **Monitoring** and alerting
- **Logging** best practices
- **Debugging** pipelines

Orchestration Tools

- **Apache Airflow** (DAG-based)
- **Prefect** (workflow engine)
- **Dagster** (data orchestrator)
- **Luigi** (Python-based)

Monitoring Tools

- **CloudWatch** (AWS)
- **Stackdriver** (GCP)
- **Azure Monitor** (Microsoft)
- **Grafana** (visualization)
- **Prometheus** (metrics)

8 Data Governance, Security & Compliance

Essential for enterprise-grade pipelines

Skills & Concepts

- **Data lineage** & cataloging
- **Data quality** rules
- **Access control** & encryption
- **Compliance:** GDPR, HIPAA, SOC2

Data Catalog Tools

- **Apache Atlas** (metadata)
- **Amundsen** (data discovery)
- **Alation** (data governance)
- **DataHub** (metadata platform)

Security & Validation

- **Great Expectations** (validation)
- **IAM** (identity management)
- **AWS IAM** (access control)
- **Azure AD** (authentication)

9 Advanced Topics & Suggested Approach

Cutting-edge Data Engineering and your learning path

Advanced Topics

- **ML Pipelines:** MLflow, TFX
- **DataOps / CI-CD:** GitHub Actions, Jenkins
- **Streaming Analytics:** Flink, Kinesis
- **Graph Data:** Neo4j, Amazon Neptune

Suggested Approach

1. **Python + SQL** + relational databases
2. Learn **ETL pipelines** using Python + Airflow
3. Move to **big data tools**: Spark, Kafka
4. Learn **cloud-native pipelines**
5. Focus on **data modeling, governance**
6. Finish with **advanced topics**